TIANYI CHEN

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EDUCATION

Johns Hopkins University, USA *September 2014 - May 2018*

- PhD in Applied Mathematics and Statistic,
  Advised by Prof. Daniel P. Robinson, and Prof. Carey E. Priebe,
  Department of Applied Mathematics and Statistic.
- Master in Computer Science,
  Advised by Prof. Jason Eisner,
  Department of Computer Science.

University of Washington, USA *August 2020 - 2023*

- Master in Applied Physics
  Department of Physics

Dalian University of Technology, China *September 2009 - May 2014*

- Bachelor of Applied Mathematics.
  Overall GPA: 4.0/4.0. Ranking: #1.

WORK EXPERIENCE

Microsoft, Redmond, USA *September 2018 – Present*

- **Principal Research Manager (Tech Lead Manager, Level 65)** *May 2025 – Present*
  Senior Research Manager / Senior Researcher (Levels 63–65) *July 2022 – May 2025*
  - Own the technical vision and roadmap for **Efficient AI** and **on-device Computer Using Agent** across the Windows org, covering on-device computer-using agents, efficient training/inference, and model compression (pruning, distillation, quantization) for large language and multimodal models.
  - **Incubation and scale-up of flagship initiatives:** Incubate seed ideas into org- and cross-org programs.
    - **Local CUA:** Microsoft's first on-device, pixel-level computer-using agent for Windows; grew from a small research incubation of 10 researchers to collaborate with 20+ partner engineers and elevated it to an org-level priority.
    - **CUA-KG (Computer-Using Agentic Knowledge Graph):** New data and knowledge infrastructure for scalable, high-success-rate agentic trajectory generation and next-generation CUA architectures; scaled from a 2-person incubation to a 10-person cross-org effort consisting of 4 teams.
    - **Efficient AI stack (e.g., OTO, WINA, DistiLLM, GETA):** Guided adoption of SOTA pruning, sparse activation, distillation, and QAT methods that deliver up to **4**× compute reduction in real products with minimal quality loss.
  - **People management and team development:** Manage and mentor researchers and engineers (e.g., senior ICs and interns such as Colby Banbury and Jongwoo Ko); provide direction and feedback, grow talent, and build an inclusive, high-trust culture that supports both deep research and reliable delivery.
  - **Cross-organizational leadership:** Establish and maintain collaborations with Windows, WinAGI, OPG, MSRA, DKI, AI Frontier, SimCity and other partner teams to align goals, secure compute and data resources, and integrate Efficient AI and Agentic AI technologies into product roadmaps.
  - **From research to product shipping impact:** Drive projects from prototypes to shipping features (e.g., Copilot AI PC scenarios, Windows Studio Effects, Teams super-resolution, Local CUA), balancing algorithmic quality, robustness, and hardware constraints (GPU/NPU).

- **Research and IP portfolio leadership:** Oversee and shape a body of work published at NeurIPS, ICLR, ICML, CVPR, ACL, etc. (including an **ICML Oral, Top 1%**, frequent social media coverage) and file **20+ patents** in efficient AI, agentic systems, and on-device intelligence.

- **Applied Scientist (Level 60–62)** *September 2018 – June 2022*
  - Primary machine learning scientist for **Office 365** and Windows experiences; researched, developed, and deployed ML models in vision, language, and multimodal settings.
  - Focused on model efficiency, robustness, and large-scale deployment, laying the technical foundation for later Efficient AI and Agentic AI initiatives.

Microsoft Research, Redmond, USA *May 2017 – August 2017*

- **Research Intern**
  Researched graph-based models and computer vision for digital inking and diagramming scenarios. Developed a generic drawing beautification engine that was later protected by US patents and adopted in Microsoft products.

Amazon, Seattle, USA *June 2016 – August 2016*

- **Applied Scientist Intern**
  Built personalized recommendation system for new-born and cold-start markets for Amazon e-commercial business, focusing on NLP-based user modeling, ranking and end-to-end system design. Collaborated with applied scientists and engineers to integrate into experimental pipelines informing product decisions.

Johns Hopkins University, Baltimore, USA *September 2014 – May 2018*

- **Research Assistant**
  Designed, analyzed, and implemented numerical optimization algorithms for sparse and nonsmooth learning, resulting in publications in SIAM Journal on Optimization and Optimization Methods and Software.

- **Teaching Assistant**
  Led TA sessions for Convexity Analysis, Optimization Algorithms, Differential Equations, Linear Algebra, and Biological Statistics.

Insilico Medicine, Baltimore, USA *June 2015 – September 2015*

- **Software Engineer and Mathematical Analyst**
  Built side-effect databases and conducted pathway-based disease analysis to support early-stage drug discovery research.

Citi Bank, Beijing, China *January 2012 – March 2012*

- **Financial Analyst**
  Analyze clients financial reports.

SELECTED RESEARCH AND PRODUCT PROGRAMS (RECENT TWO YEARS)

Local CUA (On-Device Computer-Using Agent for Windows) Feb 2025 – Jul 2025

Role: Principal Research Manager, Chief Architect & Tech Lead at Microsoft.
Scope: 10 core researchers, 20+ partner engineers across Windows and Office; elevated to org-level priority.

- Incubated and scaled **Local CUA v1–v5** from a small research prototype into Microsoft's first on-device, pixel-level computer-using agent for Windows.
- Defined the technical vision and multi-release roadmap, spanning perception, grounding, planning, and execution under strict NPU/GPU constraints.
- Coordinated cross-team staffing and resource allocation, aligning 10+ core contributors and 20+ partner engineers around shared milestones.

- Drove executive alignment through online and live demos; technical success led to project adoption as a strategic pillar for on-device Copilot scenarios.

CUA-KG: Computer-Using Agentic Knowledge Graph — Aug 2025 – Present

Role: Principal Research Manager, Chief Architect & Tech Lead at Microsoft
Scope: Scaled from 2 → 10 researchers and developers across WinAGI, OPG and ASG.

- Identified and framed the **"agentic data desert"** bottleneck for Local CUA, and initiated CUA-KG as the core data and knowledge infrastructure program.
- Led the design of a **dual-graph architecture** (Task Composition Graph and Task Execution Graph) as a shared substrate for trajectory generation and agent reasoning.
- Guided teams to achieve **70–90% successful** agentic trajectory generation across applications, significantly exceeding typical industrial and academic baselines.
- Orchestrated the development of a **CUA-KG–driven agent** (RAG-style) that reaches near–state-of-the-art performance on internal CUA benchmarks.
- Drove cross-org alignment and IP protection, resulting in multiple patent filings and adoption of CUA-KG as the planned backbone for future CUA training and evaluation.

Efficient AI Incubation and Production Portfolio — 2023 – 2025

Role: Tech Lead Manager & Research Scientist (Efficient AI) at Microsoft.
Scope: Lead research incubation and transit into productions involving 2–20 researchers and developers per effort; collaborate with Azure, Windows, Office, Teams and other research teams.

- Owned the **Efficient AI incubation** for the org, prioritizing investments in structured pruning, distillation, quantization, and LLM inference acceleration for LLMs and MMMs.
- Incubated and oversaw key algorithmic lines such as **OTO (structured pruning)**, **GETA (QAT)**, **DistiLLM** and **WINA (sparse activation)**, from early research to production shipment.
- Coordinated research and product teams to integrate these methods into shipping systems, delivering **1×–4× compute reduction with no measurable quality degradation** across:
  - Azure Cognitive Services Text-to-Speech and ZipCode Audio Transformer.
  - On-device super-resolution and video codec pipelines in Microsoft Teams.
  - Local CUA and other Copilot AI PC scenarios.
- Mentored senior ICs and interns (e.g., DistiLLM-2 ICML Oral, Top 1%), connecting their research directions to the broader Efficient AI strategy and product needs.

SELECTED PUBLICATIONS AND PATENTS

Serve roles as [P]=(co)-primary author; [C]=corresponding author; [M]=mentored primary author; [D]=developer.

Efficient AI & ML System: OTO, DistiLLM, LoRAShear, GETA, WINA, FORA.

[1] [P/C/M] S. Chen, D. Zhao, J. Ko, C. Banbury, H. Zhuang, L. Liang, **T. Chen**, *WINA: Weight Informed Neuron Activation for Accelerating Large Language Model Inference.* **ICML Workshop**, 2025.

Media/Impact: Featured by AK (@_akhaliq, 442K followers) on X (Twitter); YouTube tutorial 72K views (channel: 502K subscribers).

[2] [C/M] J. Ko, **T. Chen**, S. Kim, T. Ding, L. Liang, I. Zharkov, S. Yun, *DistiLLM-2: A Contrastive Approach Boosts the Distillation of LLMs.* **ICML Oral (Top 1%)**, 2025.

[3] [P/C/M] X. Qu, D. Aponte, C. Banbury, D. P. Robinson, T. Ding, K. Koishida, I. Zharkov, **T. Chen**, *GETA: Automatic Joint Structured Pruning and Quantization for Efficient Neural Network Training and Compression.* **CVPR**, 2025.

[4] [P/C] Z. Wang, R. Hao, H. Zhang, **T. Chen***, *DeNoiseNAS: Revisiting Single-Level Optimization for Efficient and Stable Neural Architecture Search.* **Expert Systems with Applications**, 2025.

[5] [M] J. Ko, S. Kim, **T. Chen**, S.-Y. Yun, *DistiLLM: Towards Streamlined Distillation for Large Language Models.* **ICML**, 2024.

Media/Impact: Featured by Aran Komatsuzaki (@arankomatsuzaki, 153K followers) on X (Twitter).

[6] [M] P. Selvaraju, T. Ding, **T. Chen**, I. Zharkov, L. Liang, *Fora: Fast-forward caching in diffusion transformer acceleration.* Preprint, 2024.

[7] [P/C] **T. Chen**, L. Liang, T. Ding, I. Zharkov, *OTOv3: Automatic Deep Neural Network One-Shot Training and Compression from Pruning to Erasing Operators.* Preprint, 2023.

[8] [P/C] **T. Chen**, L. Liang, T. Ding, Z. Zhu, I. Zharkov, *OTOv2: Automatic, Generic, User-Friendly.* **ICLR**, 2023.

Media/Impact: Featured by AK (@_akhaliq, 442K followers) on X (Twitter).

[9] [P/C] **T. Chen**, T. Ding, B. Yadav, I. Zharkov, L. Liang, *LoRAShear: Efficient Large Language Model Structured Pruning and Knowledge Recovery.* Preprint, 2023.

Media/Impact: Featured as **Top #1 Huggingface Daily Paper**; YouTube tutorial 1.1K views.

[10] [P] T. Ding, **T. Chen**, H. Zhu, J. Jiang, Y. Zhong, J. Zhou, G. Wang, Z. Zhu, I. Zharkov, L. Liang, *The Efficiency Spectrum of Large Language Models: An Algorithmic Survey.* Preprint, 2023.

[11] [P/C] **T. Chen**, B. Ji, T. Ding, B. Fang, G. Wang, Z. Zhu, L. Liang, Y. Shi, S. Yi, X. Tu, *Only Train Once: A One-Shot Neural Network Training and Pruning Framework.* **NeurIPS**, 2021.

Media/Impact: Featured by AI Technology Review (Jiqizhixin), 703k followers,.

Agentic AI: CUA Knowledge Graph, Local-CUA, Phi-Ground, WinClick.

[12] [P/C] **T. Chen**, C. Luo, etl. *VLM-based GUI Understanding for Computer-Use Agents.* US Patent, 2025.

[13] [P/C] **T. Chen**, P. Cameron, K. Koishida. *Systems and Methods for Generating High-Quality Trajectory Data Using a Computer Using Agent (CUA) Knowledge Graph.* US Patent, 2025.

[14] [P/C] **T. Chen**, P. Cameron, K. Koishida. *Structured, automated system for generating large volumes of realistic, diverse, and training-ready computer using tasks.* US Patent, 2025.

[15] [P/C] **T. Chen**, C. Banbury, M. Fayyaz, R. Yan, R. Gal, S. Amizadeh, S. Abdali. *Local CUA v5, a universal agentic framework for computer using agent.* US Patent, 2025.

[16] [P/C] **T. Chen**, C. Banbury, M. Fayyaz, R. Yan, R. Gal, S. Amizadeh, S. Abdali. *Collaborative and Contrastive Framework for Enhanced Agentic Planning and Execution.* US Patent, 2025.

[17] [D] M. Zhang, Z. Xu, J. Zhu, Q. Dai, K. Qiu, Y. Yang, C. Luo, **T. Chen**, J. Wagle, T. Franklin, B. Guo. *Phi-ground tech report: Advancing perception in gui grounding.* Preprint, 2025.

Media/Impact: Featured as **Top #2 Huggingface Daily Paper**, Phi-Ground-4B model.

[18] [D/M] Z. Hui, Y. Li, C. Banbury, **T. Chen**, K. Koishida, *Winclick: Gui grounding with multimodal large language models.* **ACL Main**, 2025.

[19] [P/M] H. Zhao, **T. Chen***, Z. Wang. *On the robustness of gui grounding models against image attacks.* **CVPR Workshop**, 2025.

[20] [P] Z. Wang*, **T. Chen***, X. Wang, J Jin. *Evolution of co-operation among mobile agents with different influence.* **Physica A: Statistical Mechanics and its Applications**, 2013.

Numerical Optimization: HESSO, HSPG, OBProx-SG, FaRSA.

[21] [P/C] **T. Chen**, X. Qu, D. Aponte, C. Banbury, J. Ko, T. Ding, Y. Ma, I. Zharkov, L. Liang. *Hesso: Towards automatic efficient and user friendly any neural network training and pruning.* Preprint, 2024.

[22] [P] Y. Dai*, **T. Chen***, G. Wang, D. Robinson. *An Adaptive Half-Space Projection Method for Stochastic Optimization Problems with Group Sparse Regularization.* **TMLR**, 2023.

[23] [P] Z. Zhong*, **T. Chen***, Z. Wang. *MAT: Mixed-Strategy Game of Adversarial Training in Fine-tuning*. **IJCAI**, 2023.

[24] [P/C] **T. Chen**, T. Ding, G. Wang, and Z. Zhu. *Orthant Based Proximal Stochastic Gradient Method for ℓ_1-Regularized Optimization*. **ECML**, 2020.

[25] [P] **T. Chen**. *A Fast Reduced-Space Algorithmic Framework for Sparse Optimization*. Johns Hopkins University Thesis Library, 2018.

[26] [P] **T. Chen**, F. E. Curtis, D. P. Robinson. *FaRSA for ℓ_1-regularized convex optimization: local convergence and numerical library*. **Optimization Method and Software**. 2018.

[27] [P] **T. Chen**, F. E. Curtis, D. P. Robinson. *A reduced-space algorithm for minimizing l1-regularized convex functions*. **SIAM Journal on Optimization**, 2017.

Computer Vision, Low-Level Vision, Nerf, Text to Speech

[28] [D] K. Zhang, T. Ding, J. Jiang, **T. Chen**, I. Zharkov, V. M Patel, L. Liang. *ProCrop: Learning Aesthetic Image Cropping from Professional Compositions*. **AAAI** 2026.

[29] J. Jiang, T. Ding, K. Zhang, J. Zhou, **T. Chen**, I. Zharkov, Z. Zhu, L. Liang. *Cat-AIR: Content and Task-Aware All-in-One Image Restoration*. 2025.

[30] [D] H. Zhu, T. Ding, **T. Chen**, I. Zharkov, R. Nevatia, L. Liang. *CaesarNeRF: Calibrated Semantic Representation for Few-shot Generalizable Neural Rendering*. **ECCV**, 2024.

[31] [D] J. Zhou*, T. Ding*, **T. Chen**, J. Jiang, I. Zharkov, Z. Zhu, L. Liang. *DREAM: Diffusion Rectification and Estimation-Adaptive Models*. **CVPR** 2024.

[32] [P] G. Wang*, **T. Chen***, K. Ghasedi, T. Ding, C. Nuesmeyer, I. Zharkov, M. Kankanhalli, L. Liang. *S3Editor: A Sparse Semantic-Disentangled Self-Training Framework for Face Video Editing*. Preprint 2024.

[33] [D] T. Dang, D. Aponte, D. Tran, **T. Chen**, K. Koishida. *Zero-shot text-to-speech from continuous text streams*. 2024.

[34] [P] **T. Chen**, S. Yi, *Signature verification*. US Patent, 2022.

[35] [P] **T. Chen**, Y. Shi, S. Yi, *Spatially sparse convolutional neural networks*. US Patent, 2022.

[36] [P] **T. Chen**, G. Ganesan, B. Fang, S. Yi, *Semantic Segmentation for Stroke Classification in Inking Application*. US Patent, 2021.

[37] [D] E. Shi, A. Orazaev, **T. Chen**. *Object detection and segmentation for inking applications*. US Patent, 2020.

[38] [P/C] B. Ji, **T. Chen**. *Generative Adversarial Network for Handwritten Text*. Technical report, 2019.

[39] [P] **T. Chen**, X. Tu, R. Townsend, M. Fu. *System and method for drawing beautification*. US Patent, 2017.

Safety & Privacy AI

[40] [D] R. He, K. Tong, Z. Zeng, H. Li, **T. Chen**, H. Zhuang, *Analytic Federated Learning*. **CVPR**, 2025.

[41] [D] Y. Xiao, L. Liang, **T. Chen**, C. Wang, I. Zharkov. *Disentangled Multi-band Frequency Augmentation for Corruption Robustness*. Preprint, 2024.

[42] [D] Y. Zheng, **T. Chen**, H. Zhu, C. Yang, Q. Xuan, Z. Wang. *BufferSearch: Generating Black-Box Adversarial Texts With Lower Queries*. Preprint, 2024.

[43] [M/C] Z. Wang, Y. Zheng, H. Zhu, C. Yang, **T. Chen**, *Transferable Adversarial Examples Can Efficiently Fool Topic Models*. **Journal of Computer Security**, 2022.

SKILLS

Technical: distributed training, pre-training, post-training, LLMs, multimodal models, structured pruning, distillation, quantization, computer using agent, inference acceleration, natural language processing, computer vision, deep Learning, mathematics, Python, C++, PyTorch.